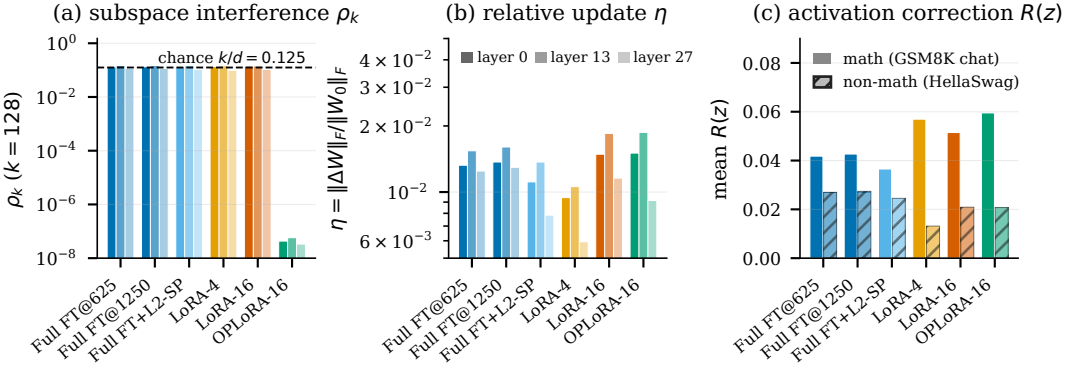
(a) subspace interference ρ_k
(b) relative update η
(c) activation correction $R(z)$
ρ_k ($k = 128$)
chance $k/d = 0.125$
$\eta = \|\Delta W\|_F / \|W_0\|_F$
layer 0
layer 13
layer 27
mean $R(z)$
math (GSM8K chat)
non-math (HellaSwag)
Full FT@625
Full FT@1250
Full FT+L2-SP
LoRA-4
LoRA-16
OPLoRA-16